

# Implement Client Script & UI Policy (Incident)

## Phase 2: Requirement Analysis – Solution Requirements

| Item          | Details                                               |
|---------------|-------------------------------------------------------|
| Project Title | Implement Client Script & UI Policy (Incident)        |
| Platform      | ServiceNow                                            |
| Target Table  | Incident [incident]                                   |
| Sub-topic     | Solution Requirements (Functional and Non-Functional) |
| Team ID       | SWTID-2026-7176                                       |

### 1. Introduction

This section translates the problem statement from the Ideation Phase into clear requirements. It states what the solution must do (functional requirements), how well it must do it (non-functional requirements), and the business rules, data elements, assumptions and constraints that shape the implementation.

### 2. Requirement Overview

Incident records must be complete and consistent before they are submitted. The solution enforces this at the user interface level using one UI Policy, one UI Policy Action and three Client Scripts on the Incident table.

#### 2.1 Business Rules

| ID  | Business Rule                                                                                          |
|-----|--------------------------------------------------------------------------------------------------------|
| BR1 | When Impact is 1 - High, an assignment group must be provided.                                         |
| BR2 | When Impact is 1 - High, Urgency must be High and cannot be edited by the user.                        |
| BR3 | When Impact is 1 - High, an incident cannot be saved unless Assigned to is filled in.                  |
| BR4 | When Impact is no longer High, the field restrictions applied by the UI Policy are reversed.           |
| BR5 | State cannot be changed by editing directly in the list view; it must be changed on the Incident form. |

### 3. Functional Requirements

| ID  | Requirement                | Description                                                                              | Priority | Rule |
|-----|----------------------------|------------------------------------------------------------------------------------------|----------|------|
| FR1 | High Impact Control policy | Create an active UI Policy on the Incident table with the condition Impact is 1 - High.  | High     | BR1  |
| FR2 | Mandatory Assignment group | A UI Policy Action on Assignment group sets Mandatory to true when the condition is met. | High     | BR1  |
| FR3 | Reverse if false           | The UI Policy reverses its actions when Impact is not High, so fields return to normal.  | High     | BR4  |
| FR4 | Read-only Urgency          | A UI Policy Action on Urgency sets Read-only to true and leaves Visible unchanged.       | High     | BR2  |

| ID   | Requirement             | Description                                                                                                         | Priority | Rule |
|------|-------------------------|---------------------------------------------------------------------------------------------------------------------|----------|------|
| FR5  | Auto-set Urgency        | An onChange Client Script on Impact sets Urgency to 1 when Impact changes to High and shows an information message. | High     | BR2  |
| FR6  | Ignore form load        | The onChange script must not run while the form is loading or when the new value is empty.                          | Medium   | BR2  |
| FR7  | Save validation         | An onSubmit Client Script blocks the save when Impact is High and Assigned to is empty.                             | High     | BR3  |
| FR8  | Error message           | The onSubmit script shows an error on Assigned to: "Assigned To is mandatory for High impact incidents."            | High     | BR3  |
| FR9  | List edit blocking      | An onCellEdit Client Script on State shows an alert and cancels the edit.                                           | Medium   | BR5  |
| FR10 | Form-based State change | Users must still be able to change State on the Incident form and save successfully.                                | Medium   | BR5  |
| FR11 | Successful save         | An incident with valid data, including Assigned to for High impact, must save without errors.                       | High     | BR3  |

## 4. Non-Functional Requirements

| ID   | Category        | Requirement                                                                                 | Applies To    |
|------|-----------------|---------------------------------------------------------------------------------------------|---------------|
| NFR1 | Usability       | Messages must be short, specific and shown next to the affected field or as a clear alert.  | FR5, FR8, FR9 |
| NFR2 | Performance     | Validation must run in the browser without noticeable delay and without extra server calls. | All scripts   |
| NFR3 | Reliability     | Rules must behave the same on every save and on every change of Impact, including reversal. | FR3, FR7      |
| NFR4 | Data Integrity  | No high-impact incident may be saved with missing ownership or inconsistent Urgency.        | FR2, FR5, FR7 |
| NFR5 | Maintainability | Each rule lives in a separately named, active configuration record with a clear purpose.    | All records   |
| NFR6 | Compatibility   | Client Scripts must use supported g_form APIs and work with isolated script mode.           | FR5, FR7, FR9 |
| NFR7 | Security        | Configuration changes require administrative or configuration access.                       | All records   |
| NFR8 | Simplicity      | The solution must stay lightweight and easy to implement in a short-duration micro project. | All           |

## 5. Data Dictionary

The Incident fields used by the solution are listed below.

| Label            | Field Name       | Type      | Use in Solution                                                                                  |
|------------------|------------------|-----------|--------------------------------------------------------------------------------------------------|
| Impact           | impact           | Choice    | Trigger for the UI Policy and the onChange and onSubmit scripts (1 = High, 2 = Medium, 3 = Low). |
| Urgency          | urgency          | Choice    | Set to 1 by onChange and made read-only by the UI Policy Action.                                 |
| Assignment group | assignment_group | Reference | Made mandatory by the UI Policy Action.                                                          |
| Assigned to      | assigned_to      | Reference | Checked by the onSubmit script.                                                                  |
| State            | state            | Choice    | Protected from list editing by the onCellEdit script.                                            |

## 6. Configuration Requirements

| Record                              | Table    | Type             | Key Settings                                                    |
|-------------------------------------|----------|------------------|-----------------------------------------------------------------|
| High Impact Control                 | Incident | UI Policy        | Active, condition Impact is 1 - High, Reverse if false, On load |
| Assignment group action             | Incident | UI Policy Action | Mandatory: True, Read only: False                               |
| Urgency action                      | Incident | UI Policy Action | Read only: True, Visible: Leave alone                           |
| Auto set urgency for high impact    | Incident | Client Script    | onChange, field Impact, Active                                  |
| Prevent save if Assigned To missing | Incident | Client Script    | onSubmit, Active                                                |
| Prevent state change via list edit  | Incident | Client Script    | onCellEdit, field State, Active                                 |

### 6.1 Save-Validation Logic

The core of FR7 and FR8 is the onSubmit check below.

```
function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
        g_form.getValue('assigned_to') == '') {
        g_form.showErrorMessage(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
        return false;
    }
    return true;
}
```

## 7. Requirement Workflow

| Step | User Action              | System Response                                |
|------|--------------------------|------------------------------------------------|
| 1    | Open a new Incident form | Form loads; UI Policy evaluates its condition. |

| Step | User Action                               | System Response                                                                                                       |
|------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 2    | Set Impact to 1 - High                    | Assignment group becomes mandatory, Urgency is set to High and becomes read-only, and an information message appears. |
| 3    | Click Submit with Assigned to empty       | Save is blocked and an error appears on Assigned to.                                                                  |
| 4    | Fill in Assigned to and click Submit      | Incident saves successfully.                                                                                          |
| 5    | Change Impact from High to Medium         | Assigned mandatory rule and read-only Urgency are reversed.                                                           |
| 6    | Double-click State in the list view       | Alert appears and the State value remains unchanged.                                                                  |
| 7    | Change State on the form and click Update | State change saves successfully.                                                                                      |

## 8. Assumptions and Constraints

### 8.1 Assumptions

- The user has administrative or configuration access to a ServiceNow instance.
- The standard Incident table and its Impact, Urgency and State choice values are in use.
- Users work on the desktop UI, where the Client Scripts are configured to run.

### 8.2 Constraints

- Client-side controls apply only to the browser form and list; they do not cover API, import or server-side updates.
- The solution is limited to the Incident table.
- Validation covers only the rules defined above.

## 9. Conclusion

The requirements define eleven functional and eight non-functional needs, tied to five business rules. Together they provide a clear basis for the design, planning and development phases that follow.